

MALA-BRANNAN, Sweden

WMO: 021470

Lat: 65.1512N

Lon: 18.5935E

Elev: 1191

StdP: 14.07

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-12.1	-22.8	1.6	-17.6	-16.6	2.3	-12.1	19.8	33.2	16.7	32.4	0.7	220	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	75.6	59.5	72.2	58.7	68.9	57.1	62.8	71.5	60.7	68.0	58.8	66.1	6.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.6	79.6	65.5	57.5	73.8	63.4	55.6	68.7	61.5	28.9	71.5	27.4	68.0	26.1	66.3	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	14.4	12.2	DB	-25.5	79.0	6.1	4.1	-29.9	81.9	-33.5	84.3	-36.9	86.6	-41.3	89.5
			WB	-25.4	64.9	6.1	2.6	-29.8	66.8	-33.4	68.3	-36.8	69.8	-41.2	71.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	13.0	14.6	22.3	33.0	43.6	51.8	58.1	54.3	45.6	33.7	23.6	18.0	(6)
	DBStd	17.81	12.72	12.89	9.67	6.87	7.06	6.34	5.60	5.74	5.86	7.42	9.81	12.33	(7)
	HDD50	6249	1147	991	858	511	224	50	3	21	155	507	791	991	(8)
	HDD65	11178	1612	1411	1323	961	662	398	224	336	583	971	1241	1456	(9)
	CDD50	561	0	0	0	0	26	104	255	153	22	1	0	0	(10)
	CDD65	16	0	0	0	0	0	2	11	3	0	0	0	0	(11)
	CDH74	178	0	0	0	0	4	29	122	23	0	0	0	0	(12)
	CDH80	19	0	0	0	0	0	1	18	0	0	0	0	0	(13)
	WSAvg	5.1	4.0	4.5	5.7	5.4	5.8	6.1	5.1	4.9	5.6	5.1	4.3	4.6	(14)
	PrecAvg	24.4	1.7	1.1	1.2	1.2	1.7	2.6	3.8	3.2	2.2	2.1	1.9	1.6	(15)
(16) Precipitation	PrecMax	32.3	3.8	2.7	2.5	2.1	3.0	4.2	6.3	7.1	5.4	3.9	4.4	3.4	(16)
	PrecMin	17.7	0.2	0.2	0.4	0.4	0.2	0.9	1.5	0.9	0.7	0.8	0.7	0.2	(17)
	PrecStd	3.8	0.7	0.6	0.6	0.6	0.8	1.0	1.3	1.8	1.3	0.9	1.1	0.7	(18)
	PrecStd	3.8	0.7	0.6	0.6	0.6	0.8	1.0	1.3	1.8	1.3	0.9	1.1	0.7	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	41.2	45.0	59.3	73.3	77.8	82.5	76.9	65.2	54.2	45.6	39.7	(19)
		MCWB	37.6	37.6	38.4	46.5	55.2	58.2	63.8	60.9	55.7	49.1	43.3	36.9	(20)
	2%	DB	36.1	38.4	41.3	52.1	67.8	72.2	77.2	73.0	60.5	48.9	41.4	36.7	(21)
		MCWB	34.2	35.3	36.0	42.0	53.1	56.7	60.3	59.8	53.5	45.4	39.6	34.6	(22)
	5%	DB	32.8	34.7	38.6	47.9	62.4	68.1	73.5	69.2	57.3	46.5	37.9	34.6	(23)
		MCWB	31.3	32.4	34.2	39.7	50.6	55.0	60.3	58.7	51.8	43.7	36.6	33.2	(24)
	10%	DB	30.0	31.5	36.1	44.4	57.3	64.7	70.2	65.2	54.8	44.1	35.4	32.6	(25)
		MCWB	29.0	29.8	32.4	37.7	47.7	53.1	58.4	57.0	50.6	41.9	34.3	31.7	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.8	38.9	46.6	57.1	60.6	66.9	64.6	57.3	49.2	43.8	37.4	(27)
		MCDB	39.8	40.7	43.9	56.9	68.4	69.8	74.6	72.7	61.1	51.8	45.4	39.3	(28)
	2%	WB	34.6	35.5	36.5	43.1	54.5	58.4	63.8	61.9	54.7	46.0	39.6	35.1	(29)
		MCDB	36.1	38.1	40.7	50.0	64.9	68.5	73.0	68.9	58.7	48.2	41.2	36.3	(30)
	5%	WB	31.5	32.6	34.7	40.4	51.3	56.5	61.9	59.9	53.0	44.1	36.7	33.3	(31)
		MCDB	32.6	34.4	38.3	46.0	61.1	65.7	70.5	66.3	56.5	46.2	37.8	34.2	(32)
	10%	WB	29.0	29.7	32.6	38.0	48.1	54.4	60.0	57.8	51.3	42.2	34.4	31.7	(33)
		MCDB	29.9	31.2	35.6	43.4	56.2	62.2	67.4	63.7	54.2	44.1	35.2	32.6	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.5	16.1	15.8	17.3	16.6	16.7	15.7	13.5	9.9	9.1	11.2	(35)
		MCDBR	10.9	10.9	14.7	21.1	26.2	22.2	22.5	21.8	16.8	11.8	8.3	10.3	(36)
		MCWBR	10.2	9.6	10.7	12.8	14.0	10.2	9.9	10.6	10.8	9.0	7.5	9.7	(37)
	5% WB	MCDBR	10.9	11.0	14.0	18.2	24.6	20.5	19.5	18.0	14.5	10.8	7.8	10.2	(38)
		MCWBR	10.3	10.0	10.7	11.6	13.9	10.2	10.0	9.9	10.0	8.7	7.3	9.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.180	0.255	0.286	0.317	0.327	0.328	0.340	0.326	0.299	0.281	0.336	N/A	(40)
	taud		1.952	2.193	2.255	2.337	2.489	2.515	2.493	2.521	2.581	2.445	2.760	N/A	(41)
	Ebn at Noon		109	205	252	272	280	282	274	268	254	200	105	N/A	(42)
	Edh at Noon		7	17	25	30	29	29	29	25	19	14	6	N/A	(43)
(44) All-Sky Solar Radiation	RadAvg		39	230	716	1283	1542	1657	1523	1155	670	264	61	11	(44)
	RadStd		3	38	59	78	182	138	168	118	84	44	11	1	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page